Note on CSA and LP-DiD methods

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Olivier Godechot
Sciences Po, CRIS, CNRS, and AxPo

Riddhi Kalsi
Sciences Po, Department of Economics, CNRS

Ulysse Lojkine
Sciences Po, CRIS, CNRS, and AxPo

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Abstract

This note clarifies the connections between the Callaway and Sant'Anna (2021) difference-in-differences estimator (CSA) and the Local Projection Difference-in-Differences estimator (LP-DiD) proposed by Dube et al. (2025). We first introduce an intermediate estimator, LP-CSA, which reformulates the CSA estimator within a local projection framework. This reformulation offers a more transparent regression representation and improves the handling of unbalanced panels.

We propose an inverse probability weighting strategy for LP-DiD that accounts for pretreatment covariate differences between treated and control units. This approach yields estimates equivalent to those of LP-CSA with inverse probability weighting and produces the equal-weighted average treatment effect estimated by CSA. It does so at a substantially lower computational cost than the regression adjustment methods proposed by Dube et al. or the covariate adjustment options available in the CSA `did` package.

Finally, we introduce the `lpdidcsa` package, which allows users to estimate various LP-DiD and LP-CSA methods.

1 CSA estimation

Callaway and Sant'Anna [2021] calculate classical difference-in-differences estimates for each treatment cohort t and each time horizon h, while ensuring that the control group consists only of never-treated or not-yet-treated units. These cohort estimates are then aggregated based on the weights of each treated cohort. Importantly, in their implementation, Callaway and Sant'Anna estimate means by "groups" (e.g., treated or control). This approach thus corresponds to a group-level fixed effects approach rather than a unit fixed effects approach.

When the panel is balanced, results are the same. However, when the panel is unbalanced, this can generate substantial differences.

The difference-in-differences estimate β_t^h at time horizon h for a cohort treated in t is as follows:

$$\beta_t^h = \left(\overline{Y_{t+h}(1)} - \overline{Y_{t-1}(1)}\right) - \left(\overline{Y_{t+h}(0)} - \overline{Y_{t-1}(0)}\right) \tag{1}$$

where Y is the average outcome for the treated (1) or control (0) group in $t+h$ or in $t-1$.

Reformulated in a regression framework, the cohort difference-in-differences estimate β_t^h can be written as follows:

$$y_{s,i} = \alpha_t^h + \delta_t^h \left(\mathbb{1}_{s=t+h}\right) + \psi_t^h \Delta T_{t,i} + \beta_t^h \left(\mathbb{1}_{s=t+h} \times \Delta T_{t,i}\right) + u_{s,i} \tag{2}$$

for period $s \in \{t-1, t+h\}$, where $\Delta T_{t,i}$ is a dummy variable indicating the time period where a unit switches from non-treated to treated: $\Delta T_{t,i} = \text{Treated}_{t,i} - \text{Treated}_{t-1,i}$.

2 LP-CSA estimation without control variables

In a balanced panel, where we observe the same units before and during treatment, the difference-in-differences estimate for a treatment cohort t and a time horizon h can be rewritten as a first-difference equation:

$$\begin{aligned} \beta_t^h &= \left(\overline{Y_{t+h,i}(1) - Y_{t-1,i}(1)}\right) - \left(\overline{Y_{t+h,i}(0) - Y_{t-1,i}(0)}\right) \\ &= \overline{\Delta_{t-1}^{t+h} Y_{p,i}(1)} - \overline{\Delta_{t-1}^{t+h} Y_{p,i}(0)} \end{aligned} \tag{3}$$

which can be rewritten in a regression style as:

$$(y_{t+h,i} - y_{t-1,i}) = \beta_t^h \Delta T_{t,i} + u_i \tag{4}$$

One of the innovations of Dube et al. [2025] is to propose a redesign of the database that allows computing these cohort-time horizon DiD estimates in a limited number of regressions. They suggest designing a clean database consisting of treated cases ($\Delta T_{t,i} = 1$) and clean controls ($T_{t+h\times(h>0)} = 0$). By excluding already treated cases, this design avoids forbidden comparisons between treated units and already treated units.

Before explaining the LP-DiD method in more depth in the next section, we would like to highlight a seemingly trivial but important consequence of this data redesign. This design allows estimating, for a given time horizon h, all the β_h^t parameters with just one simple first-difference regression. We will call this estimation the LP-CSA.

$$(y_{t+h,i} - y_{t-1,i}) = \underbrace{\sum_s \beta_s^h \left(\mathbb{1}_{s=t} \times \Delta T_{s,i}\right)}_{\substack{\text{Treatment dummy} \\ \text{interacted with dates}}} + \underbrace{\delta_t^h}_{\text{Date FE}} + u_{t,i}^h \tag{5}$$

The average β^h estimate can be obtained through a weighted mean of the regression's β_t^h parameter, following an aggregation similar to the one implemented in Sun and Abraham's approach (using the aggregate function available in the R-fixest package):

$$\beta^h = \frac{\sum_s n_s \beta_s^h}{\sum_s n_s} \tag{6}$$

where $n_t = \sum_i \Delta T_{t,i}$ is the number of newly treated units at time t.

This apparent Local Projection reformulation of CSA estimation might not seem worth mentioning. Nonetheless, it has several interesting implications.

It allows presenting CSA's estimates within a user-friendly regression framework. When the sample size is not too large, it is even possible to stack the different $(t+h)$ to $(t-1)$ first-differences in order to estimate all parameters in just one regression, thereby allowing the computation of cluster-robust standard errors that account for residual correlation for different values of h.

$$(y_{t+h,i} - y_{t-1,i}) = \sum_h \left[\sum_s \beta_s^h \left(\mathbb{1}_{s=t} \times \Delta T_{t,i}\right) + \delta_t^h \right] + u_{h,t,i} \tag{7}$$

It offers a comparison benchmark for Dube et al. [2025]'s LP-DiD estimation, which we will present in the next section.

A different approach to panel balancing

LP-CSA (and LP-DiD below) provides better handling of missing observations and attrition. Indeed, the first difference imposes a partial rebalancing on each $(t+h)$ to $(t-1)$ first-differences. However, it does not impose this rebalancing to be the same for the different time horizons h. In panels of workers, where workers regularly enter or exit the panel, imposing a rebalancing (as suggested by CSA) could lead to selecting very specific types of workers and potentially misrepresenting the phenomenon studied, as well as losing statistical power. When the variance of unit fixed effects is large, unit fixed effects implied by our design do a better job of handling missing observations than CSA's group fixed effects approach. Indeed, estimates obtained on unbalanced and rebalanced panels are often much closer with unit fixed effects than with group fixed effects.

It also allows considering a different rebalancing than the classical period rebalancing, where all observations are present between two dates, for example, 2000 and 2015. Indeed, when events are staggered, treated units will not have the same amount of pretreatment and post-treatment periods. For large h, $t+h$ estimates will mostly represent those treated at the beginning of the period, and $t-h$ those treated at the end of the period. Moreover, if the treatment can be repeated, this type of balancing does not guarantee that units at the beginning of the period are not yet treated. To address this type of concern, another rebalancing might be more informative and useful. We can impose all units to be present for a given span before and after t (e.g., 5 years before and 5 years after), with a variable t.

While it is theoretically possible to balance the panel in that way and then apply the CSA regressions on it, the data format required by CSA's "did" package makes it impractical. In contrast, the format of the data required for the LP approach naturally lends itself to it. The LP-CSA and LP-DiD frameworks allow this type of analysis.

3 LP-DiD estimation without control variables

With LP-DiD, Dube et al. [2025] make a second contribution. They developed a more efficient way of estimating β^h, which is an important contribution for large datasets.

$$(y_{t+h,i} - y_{t-1,i}) = \underbrace{\beta^h \Delta T_{t,i}}_{\text{Treatment dummy}} + \underbrace{\delta_t^h}_{\text{Date FE}} + u_{t,i}^h \tag{8}$$

for $h = -Q$ to $h = +H$ and $h \neq -1$, and where $\Delta T_{t,i} = 1$ (treated) or $T_{t+h} = 0$ (clean controls).

Note that this is the same as equation 5, except that here the treatment dummy is not interacted with the date.

Dube et al. [2025] show that equation 8 enables obtaining a "variance-weighted ATT" with

$$\beta^h_{LP-DiD} = \sum_t \omega_t^h \beta_t^h \tag{9}$$

They show that the weights ω_t^h are all positive. This means that the LP-DiD estimator avoids the TWFE "negative weights" problem resulting from "forbidden comparisons". It is generally close, but not equal, to CSA's "equal-weighted ATT" estimator discussed above:

$$\beta^h_{CSA} = \frac{\sum_t n_t \beta^h_t}{\sum_t n_t} \tag{10}$$

However, a simple reweighting of the LP-DiD enables estimating a β^h equivalent to Callaway and Sant'Anna's β^h_{CSA}.

To obtain this reweighting, they run a first-step OLS linear probability regression to predict the probability of being treated by date of treatment:

$$\Delta T_{t,i} = \sum_s \delta_s + \varepsilon_{i,t} \tag{11}$$

The residual $\varepsilon_{i,t}$ depends only on the date and the treatment status of the unit. Let us denote ε_t its value for units newly treated at date t, and note (following Dube et al. [2025], Online Appendix, p. 6) that it is equal to the share of the clean control units in cohort t, or one minus the share of newly treated units.

These residuals are then used to compute weights:

$$w_t = \frac{\sum_{s \neq 0} n_s \varepsilon_s}{\varepsilon_t} \tag{12}$$

Using this weight in equation 8 yields the equal-weighted ATT estimator β^h_{CSA}. (Alternatively, using the inverse probability weighting also generates the equal-weighted ATT estimator.)

4 LP-DiD with control variables

It would be tempting to introduce, in equation 8, some pretreatment control variables to control for differences between treated and non-treated units. LP-DiD allows doing so:

$$(y_{t+h,i} - y_{t-1,i}) = \underbrace{\beta^h \Delta T_{t,i}}_{\substack{\text{Treatment}\\ \text{dummy}}} + \underbrace{\delta^h_t}_{\text{Date FE}} + \underbrace{\sum_k \psi^h_k X_{k,i}}_{\substack{\text{Pretreatment}\\ \text{control variables}}} + u^h_{t,i} \tag{13}$$

for $h = -Q$ to $h = +H$ and $h \neq -1$.

This estimation is computationally very efficient. However, Dube et al. [2025] warn that only "under more restrictive assumptions, specifically, that treatment effects do not vary based on the value of the covariates, estimation of a simple LP-DiD specification with control variables yields a convex variance-weighted effect."

5 LP-DiD with adjusted regression

To obtain the equal-weighted ATT effect without relying on this restrictive assumption, they thus propose to interact treatment dummies with both date fixed effects and control variables. One can post-estimate β^h using the regression adjustment technique:

$$(y_{t+h,i} - y_{t-1,i}) = \sum_s \gamma_s^h \left(\mathbb{1}_{s=t} \star \Delta T_{t,i}\right) + \sum_s \sum_k \psi_{s,k}^h \left(\mathbb{1}_{s=t} \star X_{k,i}\right) + u_{t,i}^h \quad (14)$$

(We use the star symbol to indicate that the main effects are also included: $X \star Y = X + Y + X \times Y$.)

Comparison with CSA

This estimate is close but not exactly similar to the one calculated with the outcome control variable option in the Callaway's and Sant'Anna's "did" package (est_method = "reg"). To obtain CSA estimates, one has instead to compute a triple interaction between the three types of variables as follows:

$$(y_{t+h,i} - y_{t-1,i}) = \sum_s \sum_k \psi_{s,k}^h \left(\mathbb{1}_{s=t} \star \Delta T_{t,i} \star X_{k,i}\right) + u_{t,i}^h \quad (15)$$

and retrieve the β^h using the regression adjustment technique.

Efficacy

However, the latter estimation is computationally costly and often crashes when using a large number of control variables (especially with R's avg_comparison function for regression adjustment). Absorbing fixed effects does not allow for the correct estimation of standard errors. Hence, with this option, the fundamental computational advantage of LP-DiD falls short.

We therefore propose, in the next section, an alternative for efficiently estimating equally weighted ATT.

6 LP-DiD with inverse probability weighting

We propose using inverse probability weighting, also called propensity score weighting, which is widely used in the DiD literature since Abadie [2005]. This technique modifies the weights of the control group in order to make control units as similar as possible to treated units.

To do so, we first estimate the probability of treatment with a logistic regression:

$$\operatorname{logit}\left(P\left(\Delta T_{t,i}=1\right)\right)=\underbrace{\delta_t}_{\text{Date FE}}+\underbrace{\sum_k \psi_k X_{k,i}}_{\substack{\text{Pretreatment}\\ \text{control variables}}}+u_{t,i} \tag{16}$$

for i present both in $t-1$ and $t+h$, either treated in t or clean control in $t+h$.

This enables obtaining the following weights:

$$w_{i,h}^{\text{ipw}}=\begin{cases}1, & \text{if } \Delta T_{t,i}=1 \text{ (treated)}\\ \frac{\hat{p}_i}{1-\hat{p}_i}, & \text{if } \Delta T_{t,i}=0 \text{ and } T_{t+h,i}=0 \text{ (clean control)}\end{cases} \tag{17}$$

where $\hat{p}_i=P^*(\Delta T_{t,i}=1)$, the predicted probability of treatment.

Combining inverse probability weighting with re-weighted LP-DiD corresponds to a "double" reweighting:

After obtaining w^{ipw}, we run the reweighting linear probability model (equation 11) using w^{ipw} as weights. This enables obtaining the weights w_t (equation 12). We then compute $w^*_{i,t,h}=w^{\text{ipw}}_{i,h}\times w_t$ to obtain our final weights, which we will use in the LP-DiD regressions using equation 8.

Numerical examples show that this inverse probability weighting of LP-DiD yields exactly the same estimates as using inverse probability weighting with LP-CSA.

Proposition: Re-weighted LP-DiD with inverse probability weighting enables estimating the "equal-weighted ATT"

Proof

Lemma (Row duplication). For any weighted least squares regression with non-negative rational weights $w_i \in \mathbb{Q}_+$, there exist a) an integer N such as $w_i \times N \in \mathbb{N}$ and b) an unweighted database DB$'$ constructed by duplicating each observation i exactly $w_i \times N$ times, such that OLS on DB$'$ yields identical point estimates to WLS on DB.

Step 1. Dube et al. [2025] establish that on any unweighted clean first-difference database, re-weighted LP-DiD and LP-CSA yield the same equal-weighted ATT estimator β^h_{CSA}.

Step 2. Let DB be a clean first-difference database where each observation i carries a weight $w^{\text{ipw}}_{i,h}$ as defined in equation 16. By the Lemma, WLS on DB with weights w^{ipw} is equivalent to OLS on an unweighted database DB$'$, where each observation i is duplicated $w_{i,h}^{\text{ipw}}$ times.

Step 3. DB$'$ is an unweighted clean first-difference database. By Step 1, re-weighted LP-DiD and LP-CSA are equivalent on DB$'$, and both yield the equal-weighted ATT estimator β_{CSA}^{h}.

Step 4. Since WLS on DB with weights w^{ipw} is equivalent to OLS on DB$'$ (Step 2), and since LP-CSA and re-weighted LP-DiD are equivalent on DB$'$ (Step 3), it follows that LP-CSA and re-weighted LP-DiD with weights w^{ipw} on DB are equivalent. □

Comparison with the CSA package

This estimate is close but not exactly similar to the one calculated with the inverse probability weighting option (est_method = "ipw") in Callaway's and Sant'Anna's "did" packages. To obtain CSA estimates, one has instead to compute an interaction between date variables and control variables as follows:

$$\text{logit}\left(P\left(\Delta T_{t,i}=1\right)\right)=\sum_{s}\sum_{k}\psi_{s,k}\left(\mathbb{1}_{s=t}\star X_{k,i}\right)+u_{t,i} \tag{18}$$

This solution is possible within our framework but more demanding to estimate. While estimating an inverse probability weighting based on equation 18 is robust to cases where heterogeneity of the effect of the treatment according to controls varies over time, it is much more demanding computationally than estimating it based on equation 16. The method we propose can therefore be interpreted as a relaxation of robustness, arguably light in many cases, for a significant gain in computational speed.

7 LP-CSA with outcome control variables

A simple alternative to the costly regression adjustment could be to use LP-CSA with control variables:

$$\left(y_{t+h,i}-y_{t-1,i}\right)=\underbrace{\sum_{s}\beta_{s}^{h}\left(\Delta T_{t,i}\times\mathbb{1}_{s=t}\right)}_{\substack{\text{Treatment dummies}\\ \text{interacted with dates}}}+\underbrace{\delta_{t}^{h}}_{\text{Date FE}}+\underbrace{\sum_{k}\psi_{k}^{h}X_{k,i}}_{\substack{\text{Pretreatment}\\ \text{control variables}}}+u_{t,i}^{h} \tag{19}$$

for $h=-Q$ to $h=+H$ and $h\neq-1$.

And to retrieve average β^h estimates with the parameter aggregation procedure described above.

However, this estimation enables obtaining the equal-weighted average treatment effect only under the assumption that control variable effects do not vary based on the value of the covariates.

Comparison with Dube et al.'s regression adjustment framework and CSA's "reg" approach

This estimation generates results that differ both from the adjusted regression proposed by Dube et al. and from CSA with the "reg" option. The adjusted regression framework of Dube et al. adds interactions between treatment dummies and pretreatment control variables (Equation 14). CSA's reg approach comes with a triple interaction between treatment dummies, treatment cohorts, and pretreatment control variables (Equation 15). Thus, both CSA and Dube et al.'s estimation enable heterogeneous treatment effects.

However, when treatment dummies are interacted with other variables than treatment cohort, it is generally not possible to obtain β^h through a simple weighted average of regression parameters (aggregation à la Sun and Abraham). (Yet, regression adjustment based on equation 19 generates the same results as the ones we propose.)

8 The lpdidcsa R package

The lpdidcsa R package enables an estimation of the different methods discussed in this methodological note.

Six methods are available in the `lpdidcsa` function:

- `lpdid`: Basic LP-DiD method without reweighting. In the absence of control variables, it yields variance-weighted ATT estimates. With control variables, it produces variance-weighted ATT estimates, provided the effects of the control variables are homogeneous. This is the fastest method.

- `lpdid_rw`: Reweighted LP-DiD method. In the absence of control variables, it yields equally weighted ATT estimates. With control variables, it produces equally weighted ATT estimates, provided the effects of the control variables are homogeneous. This is the second fastest method.

- `lpdid_adj`: Adjusted regression LP-DiD. It yields equally weighted ATT estimates but is very slow.

- `lpdid_ipw`: Inverse probability weighting LP-DiD. Without control variables, it is equivalent to `lpcsa_rw`. With control variables, it yields equally weighted ATT estimates and is reasonably fast.

- `lpcsa`: LP-CSA. In the absence of control variables, it yields equally weighted ATT estimates. Although slower than `lpdid_rw`, it has the advantage of estimating all cohort-specific treatment effects, which can be useful before aggregating them into an average treatment effect. With control variables, however, it estimates equal treatment effects only under the control variable homogeneity hypothesis.

- `lpcsa_ipw`: Inverse probability weighting LP-CSA. Without control variables, it is equivalent to `lpcsa`. With control variables, it yields equally weighted ATT estimates. Although slower than `lpdid_ipw`, it has the advantage of estimating all cohort-specific treatment effects, which can be useful before aggregating them into an average treatment effect.

In addition to these six methods, the `lpdidcsa_data` function prepares the data and produces two types of formats. Horizon-specific variables can be designed in either a "wide" format (default) or a "long" format. The long format generates a significantly larger database in terms of memory. Its main advantage is that it allows estimation of all horizons in a single regression (option `one_reg=TRUE`). While this option is much more time-consuming and feasible only on smaller databases, it allows for global clustering of standard errors, accounting for the autocorrelation of errors between horizons.

Comparing speed of execution

To test the speed of the various methods, we generated a balanced panel dataset of 100,000 observations, consisting of 5,000 individuals observed over 20 years. Ten percent of the individuals are treated, with the treatment staggered between year 2 and year 19. We estimated treatment effects for all possible horizons, from $h = -18$ to $h = +18$. The control variables consist of a single dummy variable. The benchmark script can be found here.

Table 1 summarizes the results of the comparison. Additionally, we compare these results with those from CSA's `did` package using the same database and control variables. However, it is important to note that CSA's `did` package automatically interacts control variables with time variables, whereas our package does not do so by default.

References

Alberto Abadie. Semiparametric difference-in-differences estimators. *The review of economic studies*, 72(1):1–19, 2005.

Brantly Callaway and Pedro HC Sant'Anna. Difference-in-differences with multiple time periods. *Journal of econometrics*, 225(2):200–230, 2021.

Arindrajit Dube, Daniele Girardi, Òscar Jordà, and Alan M Taylor. A local projections approach to difference-in-differences. *Journal of Applied Econometrics*, 40(7):741–758, 2025.

Table 1: Performance Comparison of Different Methods

method	median_time	memory	iterations	format	one_reg
lpdid	574.56ms	1.23GB	10	wide	FALSE
lpdid_rw	889.85ms	1.67GB	10	wide	FALSE
lpdid_ipw	2.4s	4.71GB	10	wide	FALSE
lpcsa	1.61s	3.28GB	10	wide	FALSE
lpcsa_ipw	3.06s	6.41GB	10	wide	FALSE
lpdid_adj	31.4m	709 GB	2	wide	FALSE
lpdid	2.07s	7.57GB	10	long	FALSE
lpdid_rw	2.39s	8.04GB	10	long	FALSE
lpdid_ipw	4s	11.1GB	10	long	FALSE
lpcsa	3.09s	9.22GB	10	long	FALSE
lpcsa_ipw	4.73s	12.37GB	10	long	FALSE
lpdid	8s	12.8GB	3	long	TRUE
lpdid_rw	8.66s	14.42GB	3	long	TRUE
lpdid_ipw	35.77s	56.86GB	3	long	TRUE
lpcsa	Crashed R		3	long	TRUE
did:ipw	7.62s	15.32GB	10		
did:reg	5.04s	9.26GB	10		